4.1. Report Writing Guidelines

The formal lab report should be a complete presentation of your work on the experiment. It should be written for someone who has a physics background equivalent to this course, but who does not know anything about the experiment and the measurements you carried out. After reading your report, the reader should be able to repeat the experiment by its own. The purpose of the reports is to test both your analytical skills and your writing skills at communicating physical concepts.

4.1.1 Format – the organization and presentation of the report:

The report should be formatted in a way that clearly presents all the relevant information to the reader: text, equations, figures, etc. Some standard report formatting include:

- The report must be submitted as a PDF document. It is best to use the "Export" option to create a PDF file. After publishing to PDF, go over your report to verify that all is in place.
- Font: Times New Roman, Size 12. Line spacing: 1.5.
- All measured physical quantities must be written in a scientific format and in the SI unit system: Meters, Kilograms and Seconds.

The organization of the report is crucial. The reader anticipates a particular order for the report to be presented. Deviation from that order will mislead or confuse the reader.

4.1.2 Composition: the style in which the report is written.

- The style of scientific writing is definitive, concrete and fact-based. It is not poetic, literary, sarcastic or opinionated.
- Use past tense to describe measurements and present tense for discussion.
- Use simple, complete sentences. The writer should avoid redundant, irrelevant or circuitous statements. Stay on topic. Stay specific and to the point.
- Use **thesaurus** to improve your writing using synonyms, for example: https://www.thesaurus.com/
- Flow. Remember, you are telling a continuous story between the different sections. All elements of the report follow a logical order. Write in paragraph style. Do not write as a list.
- It is ok to repeat your ideas and statements, especially if they are important. The reader does not remember everything. It is better to use different phrasing when repeating an idea.

4.1.3 Content: The report should be organized as follows:

Title

(The title should describe your work in one short sentence. The title should be specific, descriptive, and emphasizes physical phenomenon. Write title best describing your work, don't copy title of the lab guide!)

Name, student number and email of student 1
Name, student number and email of student 2

Abstract

The abstract is a short summary, in 1-2 paragraphs of the work you have done, what you have studied and what are your conclusions:

- What we want to test? (goals),
- What is being measured, how and what are the found physical quantities? (exp. description)
- What are the primary results? What are the main conclusions?

Think of the process of writing the abstract as taking one or two sentences from each of your sections (an introductory sentence, a sentence stating the specific question addressed, a sentence listing your main techniques or procedures, two or three sentences describing your results, and one sentence describing your main conclusion). **It is advised to write the abstract after you write all the other sections.**

Example:

Correct (acceptable) abstract	Incorrect abstract
A simple pendulum is a system which exhibits harmonic motion where the period of motion is related to the length of the pendulum. In order to find this functional relation, we assembled a pendulum that approximates the assumptions of the simple pendulum system. We measured its period of motion as a function of length. We then derived the functional relation from the experimental measurements and compared the results to the physical model.	• Investigation of the properties of the mathematical pendulum. • We will study the analysis of the its measurement errors.

Theoretical background

The theoretical background should state the underlying physics of the experiment. What the theory predicts, what assumptions have been made. Define the quantities to be determined and how they are related to the directly measured quantities.

This section is short, up to half page!

Write a brief description of the theoretical model used to predict the experimental results using between 1-5 formulas.

1. For each equation, in a sentence tell the reader what the equation is about.
2. Use **word equation editor** to write your equations.
3. Number your equation, along the right margin, for easy reference later in the report. Don't write the equations later in the report, just write: as seen in equation XX.
4. An equation is often followed by a sentence that defines the variables in the equation.
5. If you think that it is important to show derivation of the theoretical model and other calculations, write it in the **appendix** section. This way the introduction part will flow better. In the introduction, refer to the calculations appearing in the appendix.

Formulas used for purpose of data analysis only, such as error (uncertainty) propagation, statistical equations, goodness of fit calculations, etc. can appear only in the appendix! They are not part of the theoretical background!

Equations must not appear later in the report! You only refer to the equations already appear here!

Material and methods

Explain briefly in your own words how the experiment was performed, what was measured and how. Include a simple diagram of the apparatus whenever possible.

List here the numerical values of the directly measured quantities (lengths, angles, weights, etc.), don't forget to include the uncertainty in measurement and units. You can use table to do so (please refer to the rules for writing tables down in this file).

This section is short, up to half page!

Results and Discussion

Describe your findings in an organized, structured, and logical manner. It is advised to think about the organization ahead of the actual writing. This section may be subdivided with descriptive subheadings as an aid in developing the narrative as necessary.

Display the data and the results using graphs, tables, and any visual object you find right. It is your choice how to present the data, so that the reader will be able to appreciate and understand it.

The text, figures, and tables should be arranged in a story-like logical progression, which gradually begins from the results of the raw measurement to the comparison with the theoretical ideas which where presented in the theoretical background section, then discussion of the conclusions and explanations that connect them.

- Present your measurement data (for example the position of cart as function of time). Usually using a figure.
- Discuss your results, state what you see in the graph, what you conclude from it.
- Use scientific writing for every measured/calculated quantity (for example the mass of the cart or the calculated friction coefficient), include the uncertainty (error) of measurement and its units. Measured/calculated quantities can be presented in-line, or in new line or in table (sometimes even figure).
- If the error of measurement was calculated using uncertainty propagation or curve fitting, include detailed calculation/regression results in the **Appendix** and refer to this calculation. Again, as in the introduction section, this is to keep the logical flow of the report.

  Note: An error of more than 10%, signifies that something is wrong with your measurement or your calculation. Try to find the most significant contribution to the error, discuss about the source of this error and how to it is possible to reduce it. If the error is due to wrong calculation, recalculate.
- Compare the experimental values you have found to the theoretical values using an appropriate statistical criterion. Refer to the equation used in the theoretical background by the equation's numbers. Discuss the outcome of your comparison.
- If needed, add a theoretical line to the figure, show agreement/disagreement to the theory **(instructions to figure/table presentation listed below).**

Requirement for figures:

Example to the correct way to present figures you can find in your laboratory guides.

1. Using a short sentence or two, present your figure. Tell the reader what is shown in the graph.
2. The figure should be centered in the middle of the page.
3. **Measured data is presented as discrete points (without connecting lines). The results of theoretical line are displayed as a smooth continuous line.**
4. Use legend when there is more than one data type displayed on the graph. Give appropriate names to the different data types.

5. If necessary, add uncertainty bars to the displayed data. Uncertainty bars are usually used to present error of measurement in calculated data and not for presentation of raw data.
6. Label the figures axis with appropriate titles. Include units of measurement. Make sure the labels are clear and readable.
7. Below the graph, each figure should have a descriptive title, starting with a number (Fig 1: title) for easy reference later in the report.
8. After the figure is presented, discuss your findings. Write a sentence or two on your observation from the graph. Tell the reader to which points to pay attention and what are your conclusions.
9. **If you have performed curve fitting to the data, add a graph of the data points and the fitted line. Below the figure:**
    - **Refer to the equation you used the fit to.**
    - **Write the calculated values including errors.**
    - **Goodness of the curve fitting must be presented in the appendix. Only refer to this.**
    - **Compare to theoretical values if needed.**
    - **Write down your observations and conclusions.**

Requirements for tables:

1. Using a short sentence or two, present your table. Tell the reader the information you want to deliver using the table.
2. Give title for each type of data presented in the table. Include units of measurement.
3. If the quantities in the column all have the same uncertainty, then the uncertainty can be indicated at the head of the column as well; for example, length ($\pm 0.005\ m$).
4. Below each place a descriptive title, starting with a number (Table 1: title) for easy reference later in the report.
5. **Do not split tables across pages of the report. Do not include long tables like the data tables from motion sensor – Do not add tables of raw data to the report! This data is better presented using graphs.**
6. After the table is presented, discuss your findings. Write a sentence or two on your observation from the table. Tell the reader to which points to pay attention and what are your conclusions.

Conclusions

A short section of 2-3 paragraphs which summarizes your main results and main conclusions. Restate your goal and the phenomena you have studied. Then, do not simply restate the results – explain your conclusions and interpretations of the results and discussion section.

If the experimental results correlate with the theoretical predictions, explain how you arrived at this conclusion and explain why your experiment succeeded. If the experiment does not agree with the theory, explain how you came to this conclusion and explain why the experiment has failed. Try to give logical and coherent explanation without using superficial statements to the disagreement. (Discuss the errors, their types, sources and ways of reducing them. Identify the most significant contributor to the error of the size measured.)

References

If you use additional sources of information except from the lab guide it is essential to credit published Journal articles/books/Website for work mentioned in your report.

For reference, **see the reference section in the lab guides on how to write this section**: Number your references and refer to them in the appropriate place in your report according to their numbering. Common scientific sources are textbooks or scientific journal articles. Attribution from Wikipedia are not accepted as valid scientific reference sources. With that said, in our course, it is allowed to cite websites (and also Wikipedia) but be careful with the information you find online and always double-check its accuracy.

Plagiarism, stealing and publication of another author's language, thoughts, ideas, or expressions and the representation of them as one's own original work. Plagiarism, failing to mention your sources is considered a serious offence of trust in the scientific community!

There are numerous formats for citation, depending on the publisher. Here we will use the APA format structure for citation. Here are how to refer to different sources using the APA format:

Citing a journal article:

Author, A. (Publication Year). Article title. Periodical Title, Volume(Issue), pp.-pp.

APA format example:

Hoogenboom, B. J., & Manske, R. C. (2012). How to write a scientific article. *International journal of sports physical therapy*, *7*(5), 512.

Citing a book in print:

Author, A. (Year of Publication). Title of work. Publisher City, State: Publisher.

APA format example:

Finney, J. (1970). Time and again. New York, NY: Simon and Schuster.

Citing a general website article with an author:

Author, A. (Year, Month Date of Publication). Article title. Retrieved from URL.

APA format example:

Simmons, B. (2015, January 9). The tale of two Flaccos. Retrieved from http://grantland.com/the-triangle/the-tale-of-two-flaccos/

Citing a general website article without an author:

Article title. (Year, Month Date of Publication). Retrieved from URL

APA format example:

APA Citation Guide. (2018, May 13). Retrieved from http://www.bibme.org/citation-guide/apa

Example to correct citation can be found at the laboratory guides.

Appendix

The appendix section contains:

- Equation derivation and numerical calculations
- Equations and calculations used for data analysis, such as error (uncertainty) propagation derivation and calculations, statistical equations, goodness of fit calculations etc.
- Curve fitting reports: residuals graphs, goodness of fit calculations, confidence intervals calculations, etc.

and any other information which is significant to the evaluation of your work, but it is too lengthy and technical hereby disrupts the flow of the report.